

~Master's thesis finalization~

Development of Qiskit-based Satellite-Assisted Quantum Key Distribution Framework

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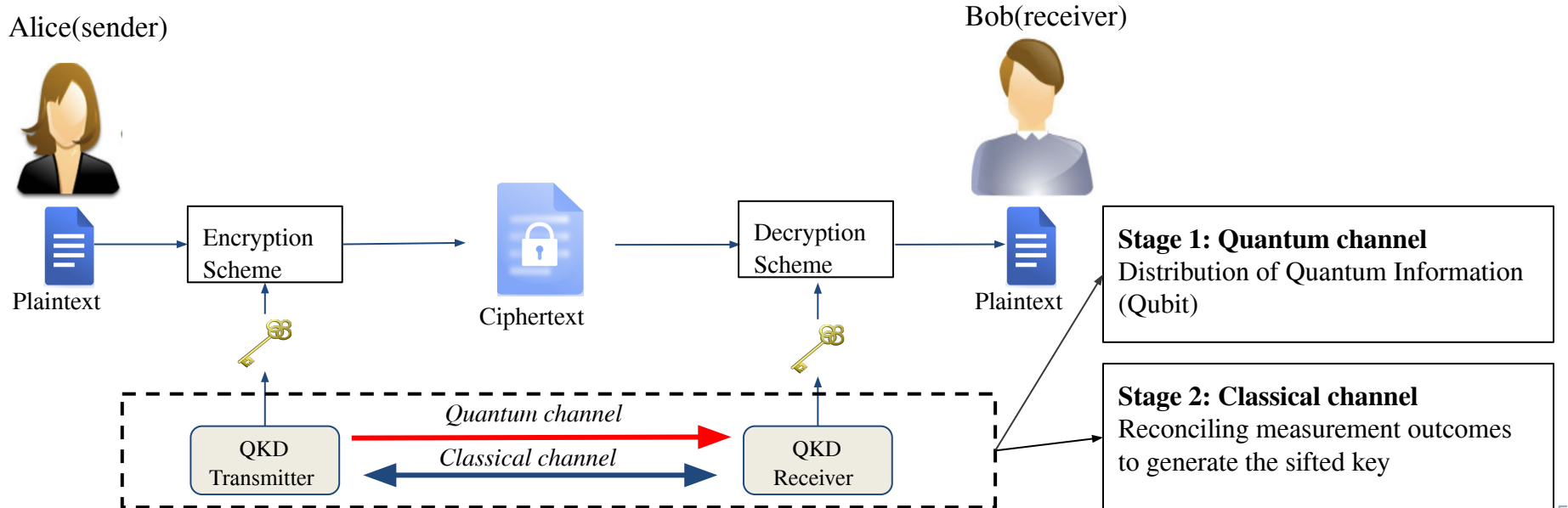
- I. Research Background
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- III. Numerical Results
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- Demand for Internet and communication networks is rapidly growing
 - Mobile/remote-user support (IoT, Internet of Vehicles, rural/oceanic areas)
 - Widespread use of cloud services and social applications
- With this rapid expansion, network security has become an increasingly critical issue
 - More connected devices = more attack surfaces
 - Sensitive data (healthcare, finance) at greater risk
- Current network security
 - Rely on **public key cryptography** (The method uses two types of keys : private key and public key)
- With advancement of Quantum Computing (Computers with unusually fast processing speeds)
 - The algorithm of complex mathematical calculations(e.g., RSA) are easily deciphered
- **To tackle this issue, Quantum Key Distribution(QKD) emerges as a promising method.**

QKD: Introduction

- QKD is a promising method to distribute secure keys secretly between legitimate users
 - It bases on the laws of quantum physics.
 - The first QKD protocol, BB84, was proposed by Bennett and Brassard in 1984.
 - Since then, several other protocols have been developed, such as E91, and BBM92.



Satellite-based QKD

QKD can be implemented over **optical fiber**, **free-space optics (FSO)-based terrestrial systems**, and **FSO-based satellite systems**.

- Challenges for conventional QKD systems
 - Optical fiber
 - Cost inflexibility and Fixed-user limitation
 - FSO-based terrestrial systems
 - Distance limitation
- For global/seamless QKD systems, **Satellite-based FSO/QKD system** have emerged as a potential architecture.
 - Flexible deployment and global coverage
 - Mobile/remote-user support (IoT, Internet of Vehicles, UAVs, rural/oceanic areas)

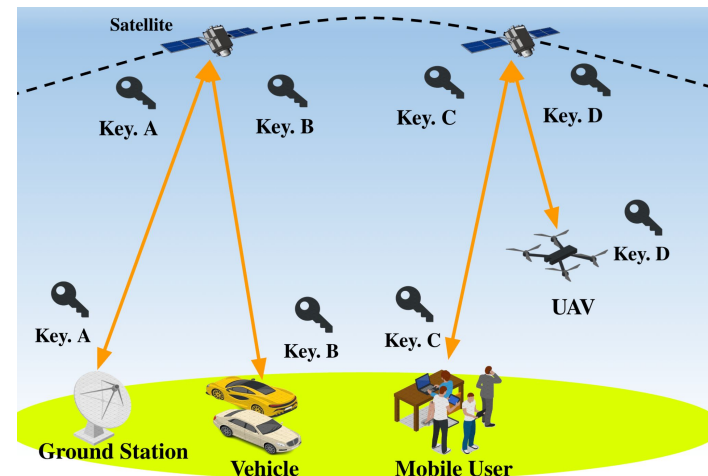


Fig. QKD services enabled by Satellite-based FSO/QKD systems.

- For the simulation & design of satellite-based QKD system, quantum computing platforms have received much attention from researcher.
- One of the well-known platforms is the IBM Quantum Experience (IQX), an open platform provided by IBM for quantum computing services.
- Recent studies have used IQX to systematically design and simulate practical QKD implementations over **optical fiber** and **terrestrial FSO systems**[3][4].

[3]: D. AL-Mubayedh, M. AL-Khalis, G. AL-Azman, M. AL-Abdali, M. Al Fosail, and N. Nagy, “Quantum cryptography on ibm qx,” in 2019 2nd International Conference on Computer Applications & Information Security (ICCAIS), 2019, pp. 1–6.

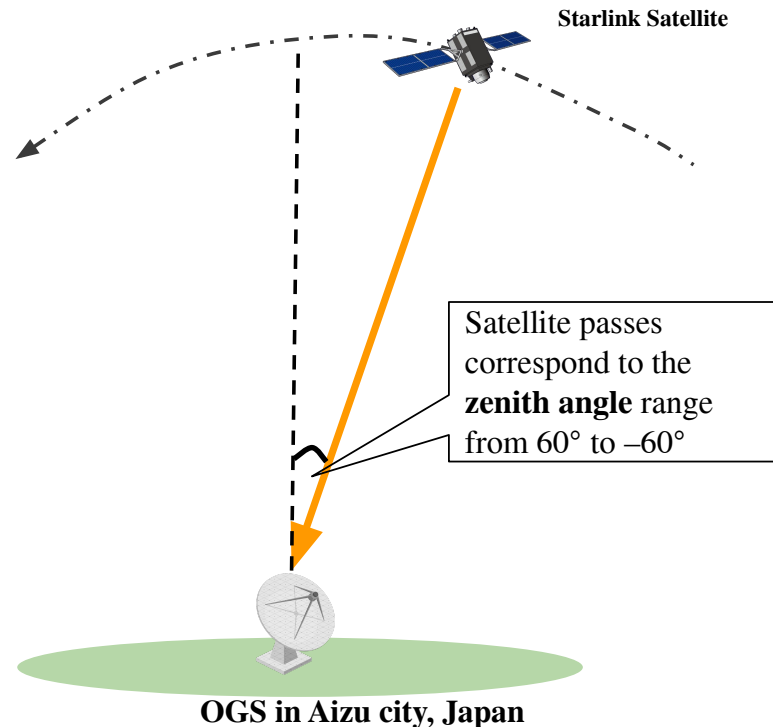
[4]: M. H. Saeed, H. Sattar, M. H. Durad, and Z. Haider, “Implementation of QKD BB84 Protocol in Qiskit,” in Proc. Int. Bhurban Conf. Appl. Sci. Technol., 2022, pp. 689–695.

- However, the Qiskit-based simulations using IQX, which incorporates quantum channel characteristics specific to **satellite-to-ground communication**, have not yet been available in the literature.

1. Develop a quantum FSO channel model for Low Earth Orbit(LEO) satellite-to-ground QKD considering major FSO impairments.
2. Implement the BB84 QKD protocol as a quantum circuit using Qiskit simulators
3. Evaluate the QKD performance under LEO satellite scenarios using key metrics

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- Scenario:
 - **Starlink LEO satellite** transmits photons to **Optical Ground Station(OGS) in Aizuwakamatsu, Japan** via a quantum FSO channel to generate a **shared secret key**.
- Case Study:
 - Starlink-1293 Satellite over Japan
- QKD Protocol Implementation
 - **Decoy-state BB84 protocol**



BB84 protocol (Basic)

- BB84 uses photon polarization states to encode the bits of the key
- Each bit is encoded with a random polarization basis: $\leftrightarrow$ (Z) or $\times$ (X)

Base \ Bit	0	1
$\leftrightarrow$ (Z)		
$\times$ (X)		



Alice(sender)

Bit	Base	Qubit State
0	$\leftrightarrow$ (Z)	
0	$\times$ (X)	
1	$\leftrightarrow$ (Z)	
1	$\times$ (X)	



Bob(receiver)

Base	Qubit State	outcome(bit)	Derived key(Sifted key)
$\leftrightarrow$ (Z)		0	0
$\times$ (X)		0	Bit is discarded
$\times$ (X)		1	Bit is discarded
$\times$ (X)		1	1

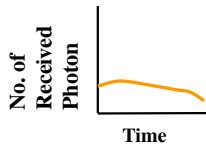
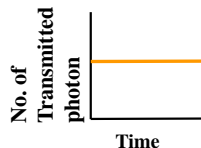
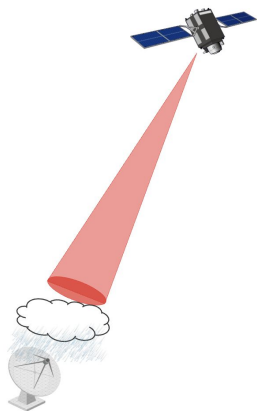
- Generated sifted key between Alice and Bob: **01**

FSO Channel Impairments

- In satellite-to-ground networks, photons are sent through the free-space optical (FSO) channel.
 - The quantum FSO channel is affected by atmospheric and alignment impairments.

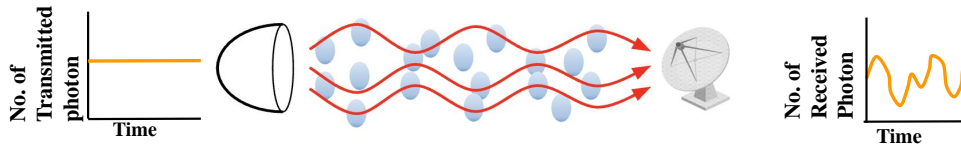
1. Atmospheric attenuation

Atmospheric absorption and scattering cause beam attenuation
→ **fewer photons reach the OGS**



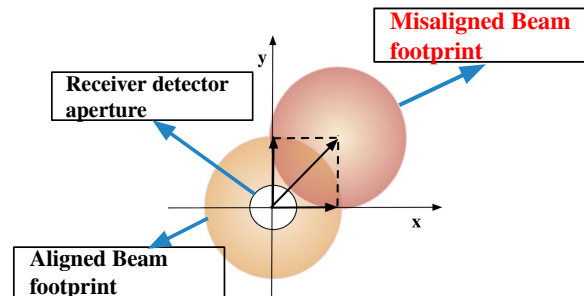
2. Atmospheric turbulence

Fluctuations in temperature and pressure change refractive index
→ beam wander & intensity fluctuations → **Fluctuating number of photons received over time**



3. Pointing error

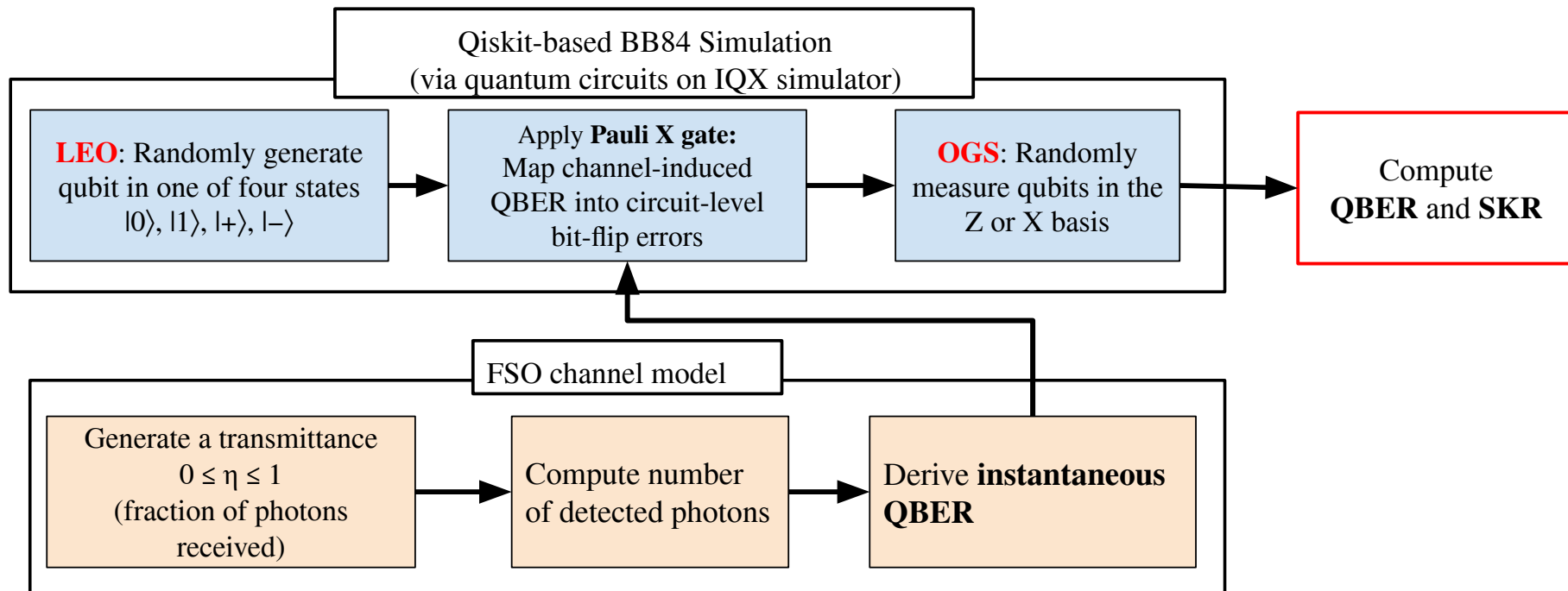
Misalignment of the beam and receiver aperture (e.g., vibration)
→ **Even small deviations cause photon loss**
→ **fewer photons received**



Qiskit-based Simulation Framework

- Qiskit-based simulation framework enables evaluation of **Quantum Bit Error Rate(QBER)** and **Secret key rate (SKR)** for QKD performance
 - **QBER**: The probability of mismatched bits in sifted key between the satellite and OGS
 - **SKR**: The number of secret key bit per second

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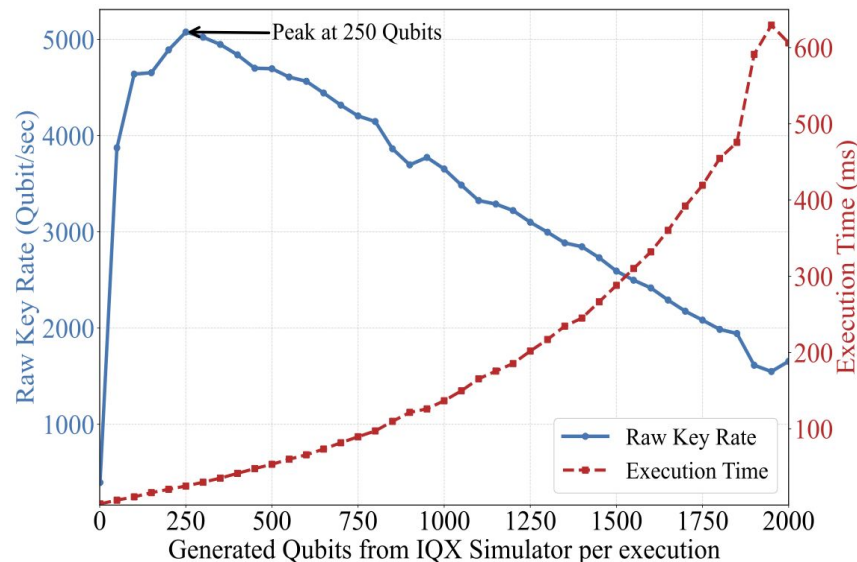


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Name	Symbol	Value
LEO Satellite		
LEO satellite altitude	H_s	550 km
Divergence half-angle	θ	$10 \mu\text{rad}$
Optical wavelength	λ	$0.85 \mu\text{m}$
Optical Ground Station		
Aperture radius	a	0.75 m
Ground station altitude	H_g	10 m
Error rate of the background	e_0	0.5
Probability of the polarization errors	e_{pol}	3.3%
Background detection probability	p_{dark}	10^{-4}
After-pulsing probability	P_{AP}	2%
Quantum Channel		
Atmospheric altitude	H_{atm}	20 km
Wind speed	v_{wind}	21 (m/s)
Mean of horizontal jitter	μ_x	0
Mean of vertical jitter	μ_y	0
Standard deviation of horizontal angle-jitter	σ_{θ_x}	$\theta/5$
Standard deviation of vertical angle-jitter	σ_{θ_y}	$\theta/5$

Numerical Results: Raw Key Rate

- ❖ Evaluated by the number of qubits generated per run on IBM Quantum Experience (IQX) and the corresponding runtime
- ❖ Simulation run on a classical computer (Linux 24.04 Workstation)



Configuration of the workstation used for simulation

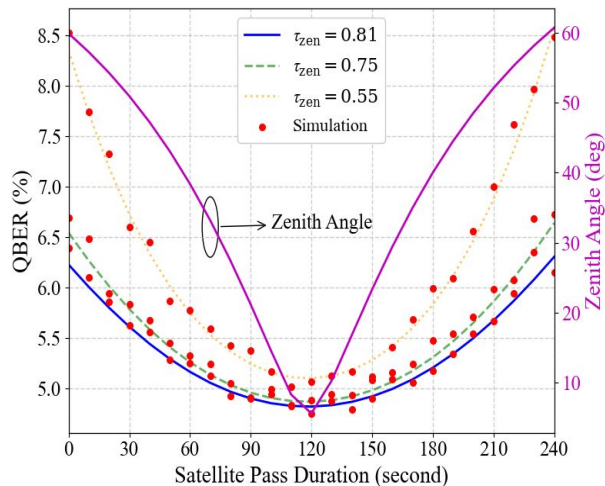
CPU Model	Intel Core i7-12700K
Cores/Threads	12 cores / 20 threads
Max Frequency	5.0 GHz

RKR performance improves with higher CPU power

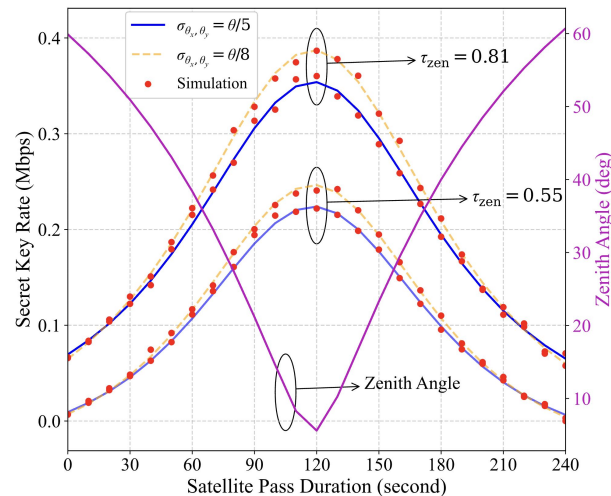
- More cores & threads
→ greater parallel processing
- Higher clock speed
(e.g., 5.0 GHz = 5 billion cycles per second)
→ faster computation

- ❖ Reaches its peak value of 5000 qubits/s when the generated qubits are 250 qubits
- ❖ Demonstrates scalability and practical feasibility of the proposed simulation

Numerical Results: QBER & SKR



Quantum Bit Error Rate over STARLINK-1293's Pass Duration



Secret Key Rate over STARLINK-1293's Pass Duration

- ❖ QBER increases with larger zenith angles due to lower transmittance and stronger turbulence.
- ❖ SKR remains higher and more stable under low jitter (yellow) than high jitter (blue).
 - High angle jitter (Blue line): $\sigma_{\theta_x}, \sigma_{\theta_y} = \theta/5$ Low angle jitter (Yellow line): $\sigma_{\theta_x}, \sigma_{\theta_y} = \theta/8$
- ❖ The theoretical results show excellent agreement with the simulation data, demonstrating the accuracy and robustness of the proposed Qiskit-based simulator.

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- Developed a Qiskit-based simulation framework for satellite-to-ground QKD
- Modeled atmospheric attenuation, turbulence, beam spreading, and misalignment over satellite networks
- Evaluated Raw Key Rate, Quantum Bit Error Rate(QBER), and Secret Key Rate(SKR) under the FSO channel conditions
- Confirmed the feasibility and scalability of the proposed framework

Thank you for your listening

Appendix: Why Decoy-state BB84?

➤ Background:

- BB84 requires single photons, but such sources are not available in practice → weak pulses are used instead.

➤ Problem:

- Weak pulses may contain multi-photon → Photon Number Splitting attack by Eavesdropper
 - He splits off one photon from a multi-photon pulse and resends the rest to OGS.

➤ Solution:

- Add decoy states(**weak** and **vacuum**)→ detect eavesdropping by comparing detection probabilities
 - Signal: normal intensity (Average no. of photon: $\mu \approx 0.5$)
 - Weak decoy: lower intensity ($\mu \approx 0.1$)
 - Vacuum decoy: no photon ($\mu = 0$)

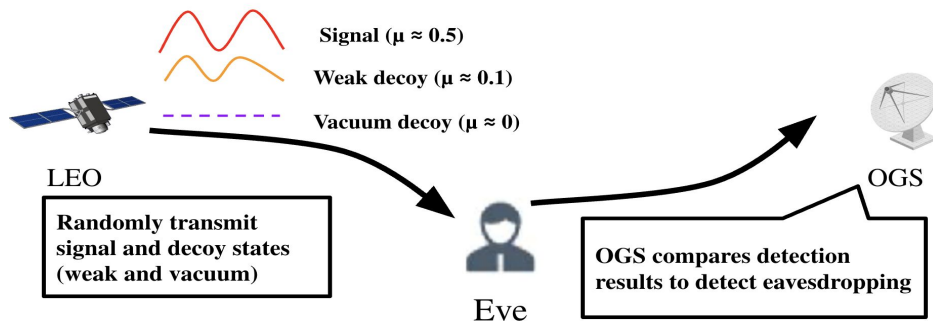


Fig. Scenario of Decoy state method

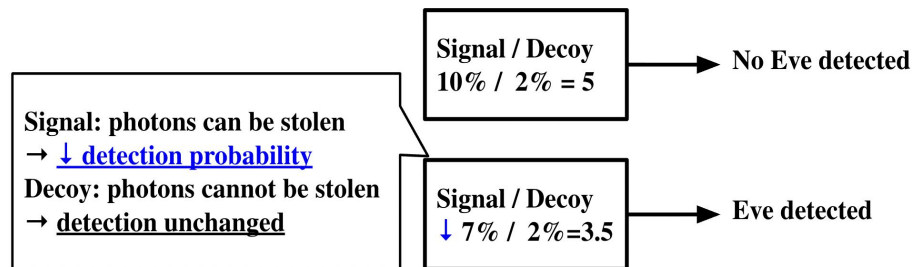
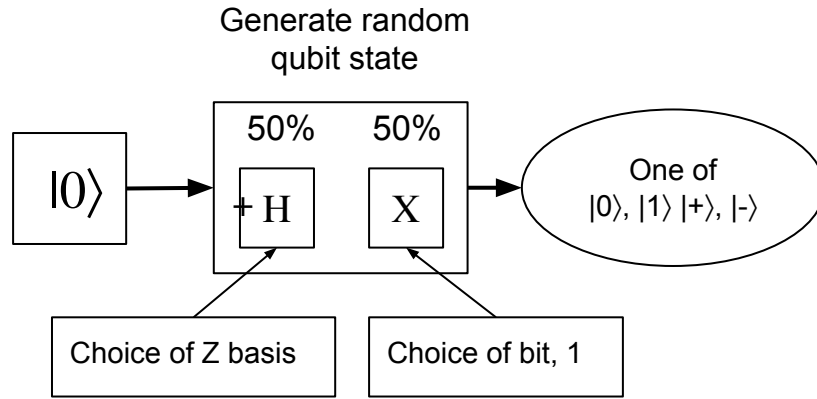


Fig. Detection Probability Ratio (Signal vs. Decoy)

Appendix: Simulation Framework

Sender's side process



Receiver's side process

